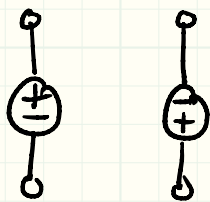


① Ohm's Law :

- Voltage (Volts)
- Resistance (Ohms)
- Current (Amps)
- $I = \frac{V}{R}$; $V = IR$; $R = \frac{V}{I}$
- Power (Watts)
- $P = I^2 R$; $P = \frac{V^2}{R}$; $P = VI$

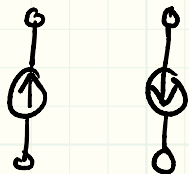
② Sources :

- Voltage Source $\rightarrow$ Constant Voltage (V)



$\rightarrow$ Series voltage sources add $\rightarrow$ KVL

- Current Sources $\rightarrow$ Constant Current (I)



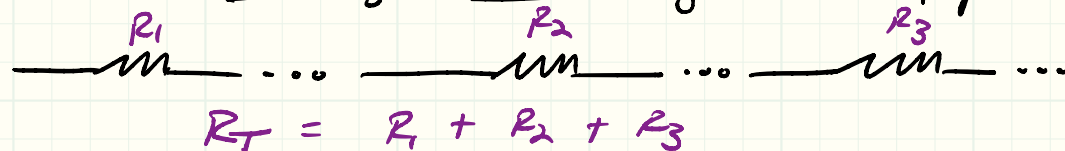
$\rightarrow$ Parallel current sources add $\rightarrow$ KCL

③ Series + Parallel Components

• Resistors :

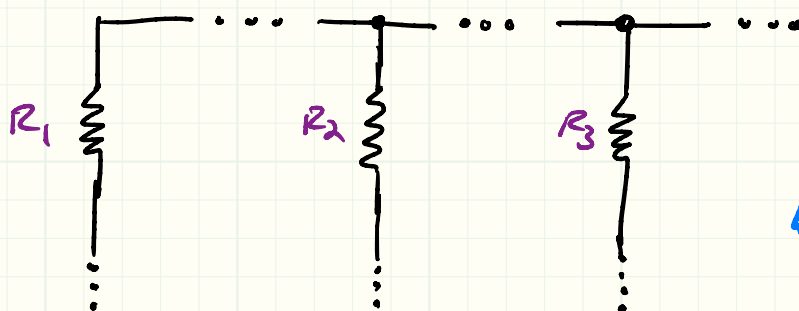
④ Series :

- R's in series just add together simply



⑤ Parallel :

- R's in parallel are "divided"



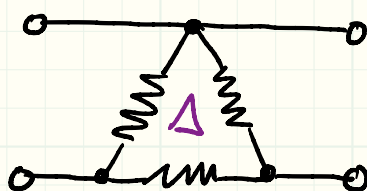
* For Two Resistors (R_1, R_2) *

$$R_p = \frac{(R_1 R_2)}{R_1 + R_2}$$

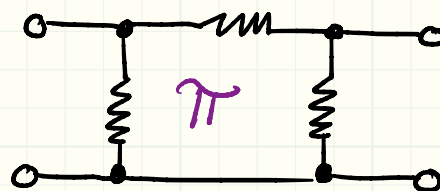
④ Delta - Wye Transforms :

- These transforms allow us to simplify complex circuits $\rightarrow$ reducing them to less resistors when possible

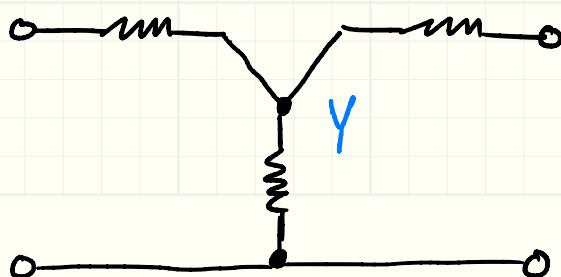
Delta :



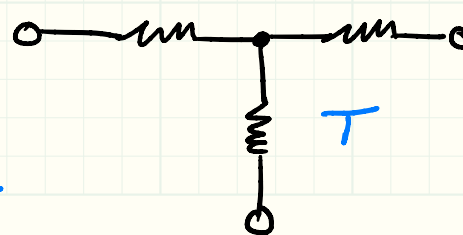
OR
 $\Delta = \pi$



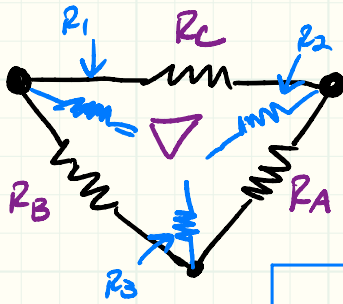
WYE :



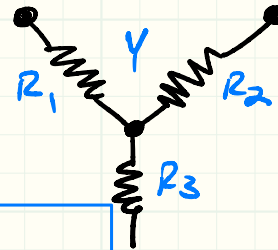
OR
 $Y = T$



Delta to Wye:



$\Delta \rightarrow Y$
to



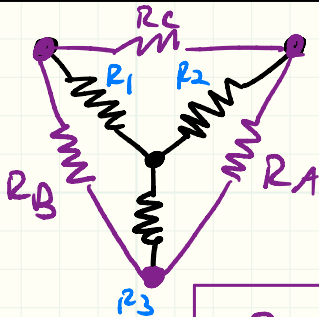
$$R_N = \frac{(R_{\text{left}})(R_{\text{right}})}{\text{sum of all } R\text{'s}}$$

$$R_1 = \frac{(R_3)(R_C)}{R_A + R_B + R_C}$$

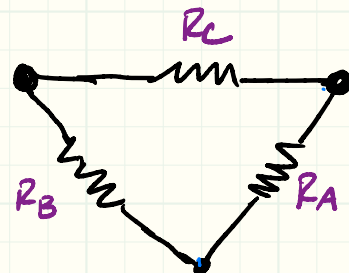
$$R_2 = \frac{(R_C)(R_A)}{R_A + R_B + R_C}$$

$$R_3 = \frac{(R_B)(R_A)}{R_A + R_B + R_C}$$

Wye to Delta:



$Y \rightarrow \Delta$
to



$$R_N = \frac{\text{Sum of multiples}}{\text{Resistor that doesn't touch node!}}$$

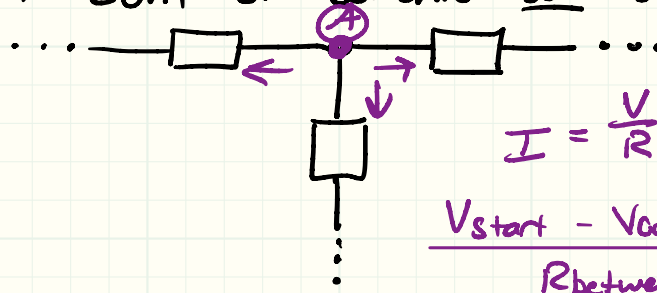
$$R_A = \frac{R_1 R_2 + R_2 R_3 + R_3 R_1}{R_1}$$

$$R_B = \frac{R_1 R_2 + R_2 R_3 + R_3 R_1}{R_2}$$

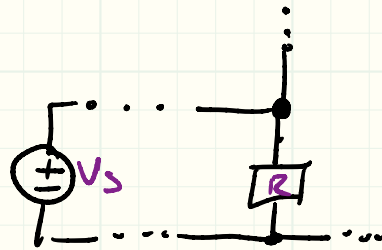
$$R_C = \frac{R_1 R_2 + R_2 R_3 + R_3 R_1}{R_3}$$

⑤ KCL and KVL (Node and Mesh):

KCL → Sum of currents out of node = zero!



$$\frac{V_{\text{start}} - V_{\text{ending}}}{R_{\text{between}}} + \dots = 0$$



If V_s touches a node without resistor between → node voltage is V_s !

Steps to Solve KCL (Node) Problems:

- Step ①: Label all nodes
- Step ②: Write KCL Eq's at all nodes
- Step ③: Plug In Given Values
- Step ④: Rewrite KCL Eq's for System of Equations
- Step ⑤: Solve the System of Equations using:
 - Ⓐ: Plug N chug
 - Ⓑ: Cramer's Rule
 - Ⓒ: Calculator * Always use to check solution!

$$V = A^{-1} * B$$

States ↑
(node voltages)
Inputs ↑
(sources)